Spotify Customer Analytics & Targeting Strategy

Using behavioural data to identify customer groups and develop data-driven targeting strategies for Spotify.

The Analytics Pipeline

01 Customer Data: Raw demographic & behavioral signals collected at scale.

02 Behaviour Analysis: Identifying patterns in engagement & preferences.

03 Segmentation: Grouping users into actionable clusters.

04 Targeting: Directing strategies to the right customer groups.

05 Personalization: Delivering tailored experiences that drive retention.

Why Customer Analytics?

Different Behaviours: Customers exhibit diverse listening habits; a single strategy cannot effectively address every user.

Uncovering Hidden Groups: Analytics reveals meaningful customer segments invisible in aggregated data.

Personalized Action: Enables targeted marketing speaking directly to segment needs.

Business Impact: Drives higher retention, engagement, and conversion via evidence-based interventions.

Core Focus: Utilize Spotify customer behavioural data to uncover distinct groups and design evidence-based marketing campaigns.

Spotify Company & Customer Ecosystem

Customer Tiers

Premium Users

Paid subscription tier providing ad-free music and offline listening. High long-term value requiring dedicated retention and upselling strategies.

Ad-Supported Users

Free access tier monetized through advertising. Acts as the primary funnel for potential Premium conversion initiatives.

Key Behavioural Signals

- **Usage Signals:** Weekly listening hours, playlist creation count, daily track skips, inactivity periods.
- **Preferences:** Recommendation satisfaction ratings, genre preferences, primary device usage.
- **Account Data:** Subscription tier status, trial interactions, ad engagement levels.

Key Strategic Insight

Spotify's continuous digital interactions generate rich behavioural signals that can be systematically analysed to decode customer needs and predict user lifetime value.

Problem Statement & Research Objectives

The Business Problem

Spotify customers differ significantly in engagement, activity levels, content preferences, and subscription status. Treating them as one homogeneous customer base limits marketing effectiveness and wastes promotional budget.

Main Research Question

"How can customer analytics identify meaningful Spotify customer segments and support more effective targeting?"

Project Objectives Pathway

1 **Analyse Behaviour:** Study listening & usage patterns.

2 **Adapted RFM:** Develop digital RFM proxy framework.

3 **Segment Users:** Perform K-Prototypes clustering.

4 **Statistical Validation:** Conduct Chi-square & Logistic regression.

5 **Visualize Insights:** Build interactive Power BI dashboard.

6 **Targeting Strategy:** Craft segment-specific action plans.

Customer Data & Analytics Framework

50,000
CUSTOMER RECORDS ANALYZED

Kaggle
SPOTIFY BEHAVIOR DATASET SOURCE

Main Dataset Variables

Category	Variable Name	Description & Context
Demographic	Age, Country	Customer age brackets and geographical locations.
Subscription	Type, Status	Free vs. Premium tier, active vs. cancelled status.
Behavioural	Hours, Playlists, Skips, Inactivity	Weekly listening hours, playlists created, daily skips, months inactive.
Experience	Recommendation Rating, Feature	1-5 scale rating for recommendations, most-liked feature.
Advertising	Ad Interaction, Ad Conversion	Interaction history with ads and converted upgrade responses.

Analytical Workflow

Dataset Acquisition → Data Preparation → RFM Behaviour Profiling → Statistical Validation → Clustering

Data Preparation & Methodology

Data Preparation Steps

1. **50,000 Records Loaded:** Imported complete Kaggle dataset.
2. **Field Cleansing:** Removed irrelevant and empty fields.
3. **Missing Values:** Verified and handled missing entries.
4. **Duplicate Removal:** Eliminated redundant records.
5. **Type Verification:** Validated numerical & categorical types.
6. **Variable Selection:** Isolated key behavioural attributes.

Analytical Toolkit

RFM Analysis Framework

Customer activity and value profiling using digital proxies adapted specifically for Spotify's streaming ecosystem.

K-Prototypes Clustering

Advanced mixed-type algorithm simultaneously handling numeric usage metrics and categorical features.

SPSS Statistical Testing

Descriptive metrics, Chi-square tests of independence, and logistic regression models.

Power BI Visualization

Interactive executive dashboards supporting business analysis and strategy formulation.

Analytical Journey: Raw Data → Clean Data → Customer Profile → Segmentation → Targeted Action

RFM-Based Customer Behaviour Analysis

The conventional transactional RFM model was adapted for Spotify's behavioural context — substituting spend and transaction counts with streaming usage proxies.

Adapted RFM Framework Dimensions

RFM Dimension	Spotify Behavioural Proxy	Strategic Rationale
R - Recency	Months Inactive / Recent Activity	Indicates how recently a customer actively engaged with the platform.
F - Frequency	Listening Hours & Usage Frequency	Reflects depth, regularity, and habitual dependence on Spotify.
M - Monetary / Value	Subscription Status & Conversion Potential	Proxies financial contribution and likelihood of premium upgrades.

RFM Scoring Logic

Recent + Frequent + High-Value: Highly engaged user with strong retention and upselling potential.

Inactive + Low Engagement: At-risk user requiring proactive win-back interventions.

Why Adapt RFM?

The dataset lacks individual monetary transaction logs. Behavioural proxies allow us to build a robust value framework without needing legacy e-commerce spend data.

Viva Voce Distinction: RFM is used here as a *profiling framework*, whereas final segmentation is executed via *K-Prototypes clustering*.

Customer Behavioural & Statistical Analysis

Behavioural Variables Analysed in SPSS

- Listening Hours
- Playlists Created
- Skips per Day
- Months Inactive

Key Statistical Modeling Results

CHI-SQUARE $X^2(3)$	LR X^2 TEST	MODEL AUC	MCFADDEN R^2
3.932	36.44	0.531	0.0019
p = 0.269 (Not Sig)	p = 0.448 (Not Sig)	Near-Random Fit	Very Limited Fit

Statistical Interpretation

No statistically significant association was found between customer segment and ad conversion. Standard demographic/behavioural traits alone had minimal predictive power for ad conversions, proving that ad response requires specialized contextual targeting.

Customer Segmentation Using Clustering

Why K-Prototypes Clustering?

While RFM provides qualitative profiling, clustering mathematically discovers natural user groupings across mixed data types:

Numerical Variables: Age, listening hours, playlists, skips, inactivity, recommendation rating.

Categorical Variables: Country, favorite genre, subscription status, primary device, favorite feature.

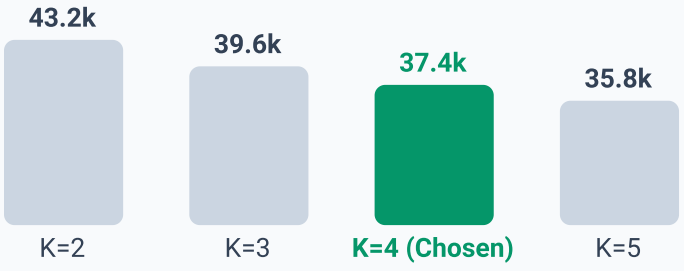
Model Fitting Process

1 **Model Fit:** Trained on 15,000 user sample records.

2 **Cluster Select:** Evaluated optimal K cost elbow curve.

3 **Assign Users:** Applied labels to full 50,000 dataset.

Cluster Cost by Number of Clusters (K)



Key Finding: Four clusters (K=4) yielded the optimal trade-off between statistical fit and actionable marketing interpretation.

Interpretation of Customer Segments

Segment 1: Young Active Listeners

Profile: Younger demographic, highly active daily listening, high recommendation satisfaction.

Opportunity: High conversion target for Premium plans and student promotional bundles.

Segment 2: Established Satisfied Listeners

Profile: Older demographic, highly consistent active usage, satisfied with ecosystem.

Opportunity: Long-term retention and upselling to Spotify Family/Duo tier plans.

Segment 3: Dissatisfied Active Users

Profile: Active listeners showing significantly lower recommendation satisfaction ratings.

Opportunity: Enhance discovery features, algorithmic fixes, and curated daily mixes.

Segment 4: Lapsed / At-Risk Listeners

Profile: 3+ months inactive, declining platform interactions across all devices.

Opportunity: Win-back campaigns, personalized push notifications, and comeback offers.

Critical Insight: The strongest segment differences are driven by *age*, *recommendation satisfaction*, and *inactivity duration* — NOT simple listening volume alone.

Project Summary & Key Takeaways

What We Did

01 **Analysed 50,000 Records:** Cleaned & structured Kaggle dataset.

02 **Applied RFM Proxies:** Created streaming-specific Recency, Frequency, Value metrics.

03 **Validated via SPSS:** Executed Chi-square & Logistic regression testing.

04 **Segmented via K-Prototypes:** Derived 4 robust customer clusters.

05 **Built Power BI Dashboard:** Formulated evidence-based targeting frameworks.

Core Findings Summary

- Age, satisfaction, and inactivity are the dominant drivers of customer variation.
- No significant statistical link exists between customer segment and ad conversion ($p = 0.269$).
- Four actionable segments were successfully established for strategic targeting.

Strategic Implication: Spotify's diverse customer base requires differentiated, segment-specific marketing — from Premium conversion for young users to win-back prompts for lapsed listeners.

Power BI Dashboard & Visualization

50,000	17,486	4,352	4
TOTAL CUSTOMERS	AD INTERACTORS	AD CONVERSIONS	SEGMENTS BUILT

Power BI Dashboard Visualization Components

Customer Distribution: Visualizing total volume share across 4 segments.

Listening Hours by Cluster: Benchmarking engagement levels per segment.

Inactivity Monitoring: Highlighting at-risk cohorts by inactivity months.

Recommendation Rating: Distribution of satisfaction scores.

Subscription Breakdown: Free vs Premium ratio per cluster.

Ad Conversion Funnel: Impression → Interaction → Conversion performance.

Interactive Dashboard Filter Controls

Slicers enabled for: **Country** | **Segment** | **Subscription Type** | **Device Type** | **Preferred Genre** | **Age Bracket**

Customer Targeting Strategy

Segment Strategic Action Plan

Segment	Primary Objective	Marketing Strategy & Campaign Tactics
Young Active	Convert to Premium	Premium free trials, student pricing messaging, social sharing features.
Established	Retain & Upsell	Family/Duo tier upgrades, loyalty perks, exclusive early access events.
Dissatisfied	Re-Engage	Improved algorithmic recommendations, guided discovery, daily mix prompts.
At-Risk	Win-Back	Re-engagement emails, "We Miss You" playlists, comeback discount offers.

Marketing Analytics Logic

1. Customer Need **Identify Trial / Loyalty Needs**

2. Objective **Set Goals: Convert or Retain**

3. Segment **Group Users by Behaviour**

4. Targeted Action **Execute Precision Offers**

Core Transition: This strategy transforms standard statistical data into true, ROI-driven *Marketing Analytics*.

Personalization & Customer Engagement

4 Dimensions of Personalization

1. Behaviour: Tailoring feeds based on explicit streaming actions.

2. Preference: Matching UI layouts to favorite features & genres.

3. Satisfaction: Intervening when recommendation ratings drop.

4. Lifecycle Stage: Serving messages aligned to user maturity.

Segment Personalization Execution

Young Active: Deliver trending viral content + trial prompts.

Established: Deliver loyalty messaging + family share invites.

Dissatisfied: Deliver genre recalibration + discovery tools.

At-Risk: Deliver favorite nostalgic content + comeback offers.

Strategic Purpose

"The objective is not simply to identify segments, but to use those segments to make customer experiences significantly more relevant."

Key Findings & Managerial Insights

5 Key Managerial Takeaways

1. **4 Distinct Profiles:** Clear behavioural clusters exist across Spotify's user base.
2. **Inactivity Signals:** Inactivity duration is the single strongest predictor of churn risk.
3. **Recommendation Satisfaction:** Drives engagement differences among active users.
4. **Listening Volume Uniformity:** Volume alone does not differentiate user segments.
5. **Ad Conversion Independence:** Segmentation does not automatically predict ad conversion.

Executive Action Direction

Customer segmentation should be deployed primarily for **personalization**, **engagement**, and **lifecycle marketing** — rather than assuming that a specific customer cluster will automatically generate higher ad conversions.

Recommendations & Business Impact

Five Core Strategic Recommendations

1. **Segment-Based Marketing:** Replace mass marketing with segment-tailored messaging.
2. **Lifecycle Marketing Automation:** Implement triggers for Convert, Retain, Engage, and Win-back workflows.
3. **Recommendation Fixes:** Prioritize algorithm refinements for users reporting low satisfaction.
4. **Proactive At-Risk Alerts:** Set automatic triggers when inactivity exceeds 30 days.
5. **Continuous Power BI Tracking:** Monitor segment size, engagement, and conversion in real-time.

Expected Business Impact Areas

- Better Marketing Resource Allocation
- Higher Retention & Premium Conversions
- Improved Platform User Engagement
- More Relevant Offer Targeting

Financial Disclaimer

Quantifying precise monetary financial impact requires access to internal Spotify user lifetime value (LTV), acquisition cost (CAC), and margin data.

Conclusion & References

Final Analytical Conclusion

This study proves how raw customer streaming data can be transformed into actionable marketing insights through adapted RFM profiling, statistical testing, K-Prototypes clustering, and Power BI visualization.

The Complete Analytics Pathway

Data Intake	Analysis	Segmentation	Activation
50,000 Customer Records	RFM & SPSS Statistical Tests	K-Prototypes 4 Clusters	Power BI Dashboard Action

> **"The real value of customer analytics is not just knowing who the customers are, but knowing what to do differently for each customer group."**

References & Data Sources

- Kaggle Dataset: *Spotify User Behavior and Pattern*
- Spotify Technology S.A. Annual Reports & Investor Relations Statements
- Project SPSS Statistical Modeling & K-Prototypes Clustering Outputs